

Python Programming

Unit 06 – Lecture 02: NumPy Functions (Stats, Sorting, Searching)

Tofik Ali

School of Computer Science, UPES Dehradun

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Repository: <https://github.com/tali7c/Python-Programming>

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Core Concepts

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Interactive

Summary

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1 Core Concepts

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Learning Outcomes

- Use important NumPy functions: `reshape`, `sum`, `mean`, `std`

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- Generate arrays using random, zeros, ones, empty

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Learning Outcomes

- Use important NumPy functions: `reshape`, `sum`, `mean`, `std`
- Generate arrays using `random`, `zeros`, `ones`, `empty`
- Sort and search arrays
- Compute dot products and matrix multiplication

Reshaping Arrays

```
a = np.arange(1, 10)
m = a.reshape((3, 3))
```

- Total elements must match (size stays same)

Statistics

■ sum, mean, std, min, max

```
print(m.sum(axis=0)) # column sums  
print(m.sum(axis=1)) # row sums
```

Statistics

- sum, mean, std, min, max
- Use axis= for row/column-wise operations

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print(m.sum(axis=1)) # row sums
```

Sorting and Searching

- `np.sort`, `np.argsort`

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- `np.max, np.argmax`

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- `np.sort, np.argsort`
- `np.max, np.argmax`
- `np.where(condition)`

Dot Product / Matrix Multiplication

```
a = np.array([1, 2, 3])  
b = np.array([4, 5, 6])  
print(a.dot(b))
```

```
M = np.arange(1, 10).reshape((3, 3))  
print(M @ M)
```

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Checkpoint 1

Question: What does `axis=0` mean in `sum(axis=0)` for a 2D array?

Checkpoint 2

Question: What is the difference between `np.sort(a)` and `a.sort()`?

Think-Pair-Share

Discuss:

- When should you use `reshape` in data analysis?

Key Takeaways

- `reshape` changes shape without changing data

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- Axis-based operations compute row/column stats

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- reshape changes shape without changing data
- Axis-based operations compute row/column stats
- NumPy supports fast sorting/searching and linear algebra operations

Exit Question

Name one NumPy function to generate a sequence of numbers.